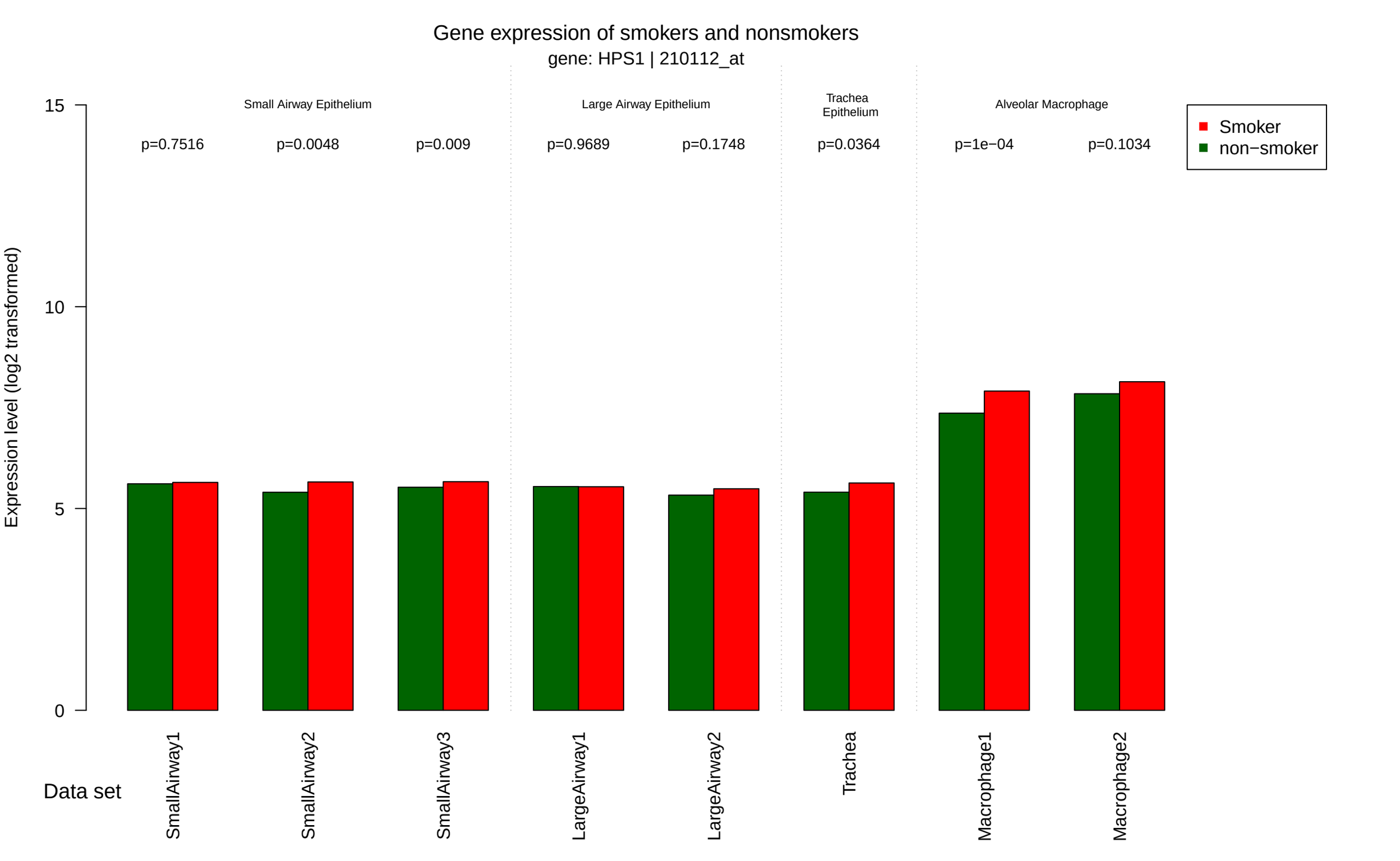
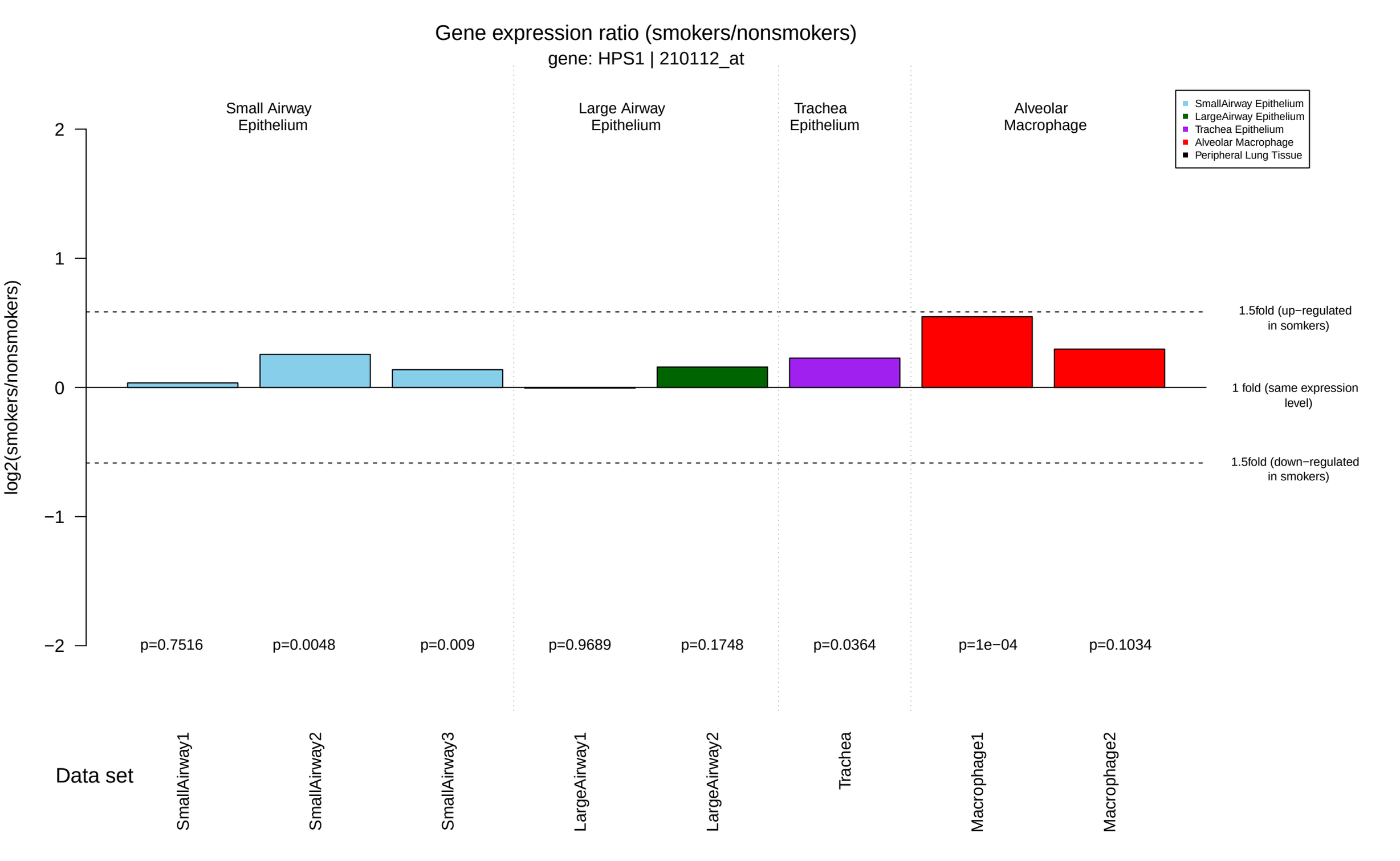
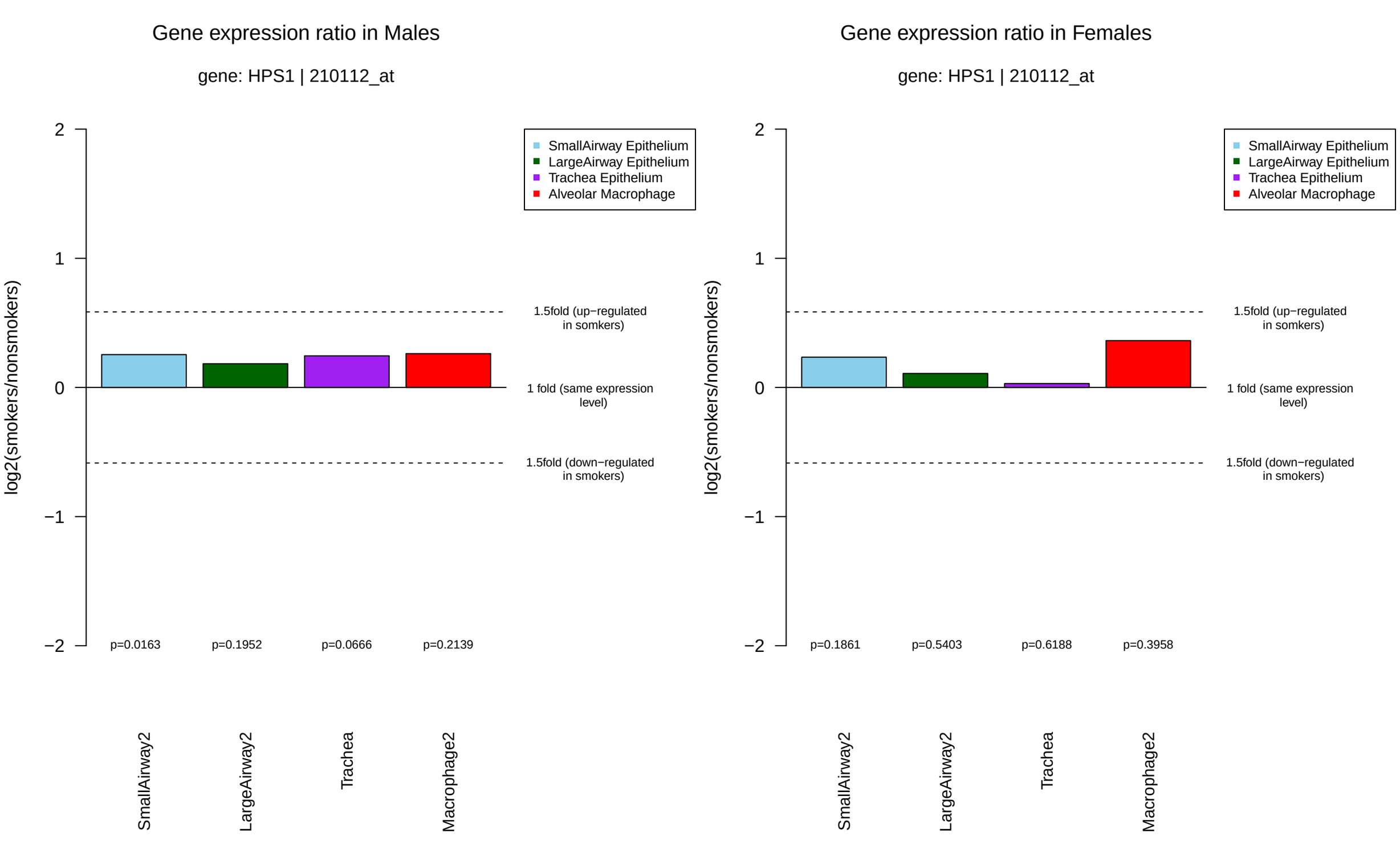
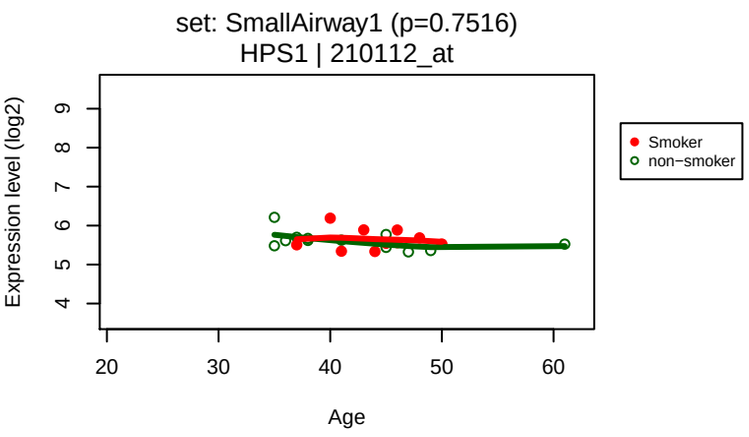
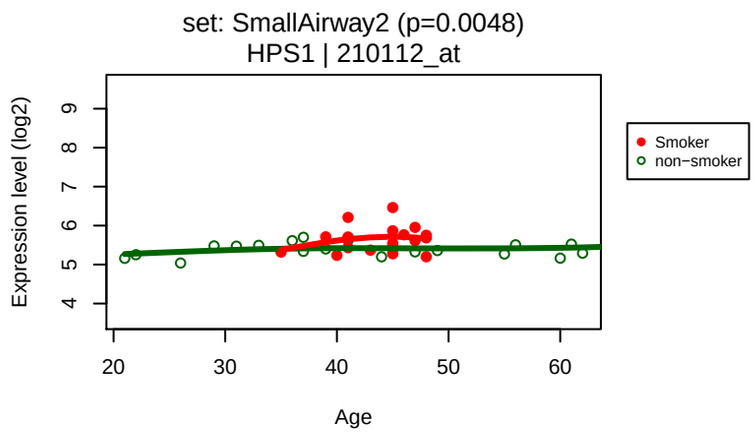
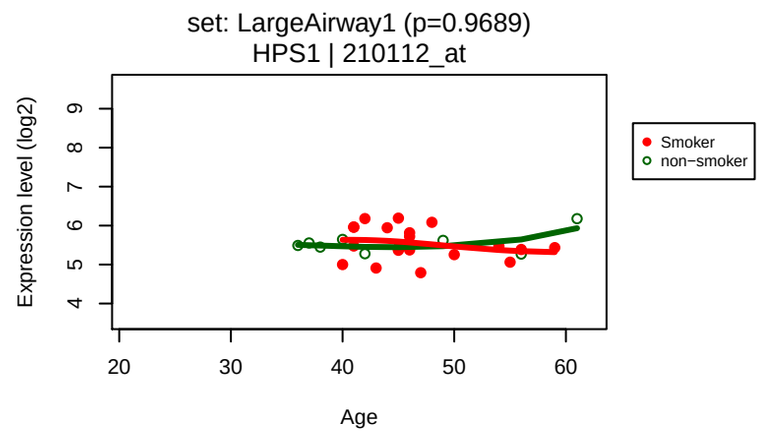

Gene expression of smokers and nonsmokers
gene: HPS1 | 210112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7516
p=0.0048
p=0.009
p=0.9689
p=0.1748
p=0.0364
p=1e−04
p=0.1034
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

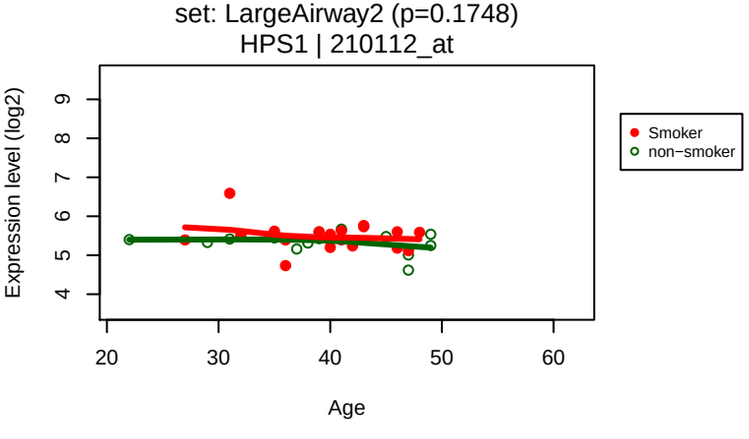

Gene expression ratio (smokers/nonsmokers)
gene: HPS1 | 210112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7516
p=0.0048
p=0.009
p=0.9689
p=0.1748
p=0.0364
p=1e−04
p=0.1034
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

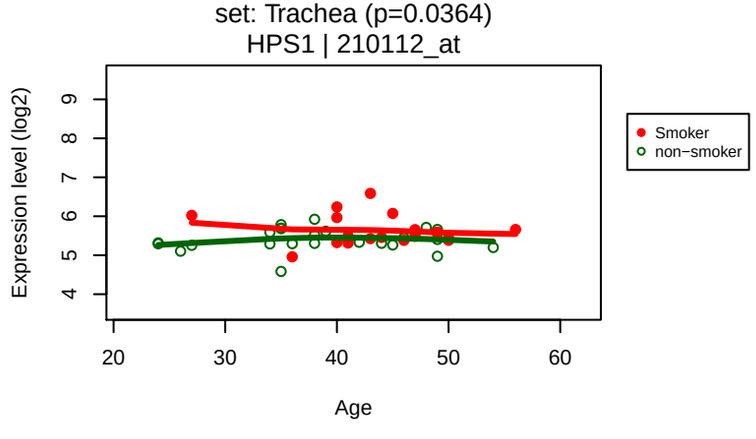

Gene expression ratio in Males
gene: HPS1 | 210112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0163
p=0.1952
p=0.0666
p=0.2139
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS1 | 210112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1861
p=0.5403
p=0.6188
p=0.3958
SmallAirway2
LargeAirway2
Trachea
Macrophage2

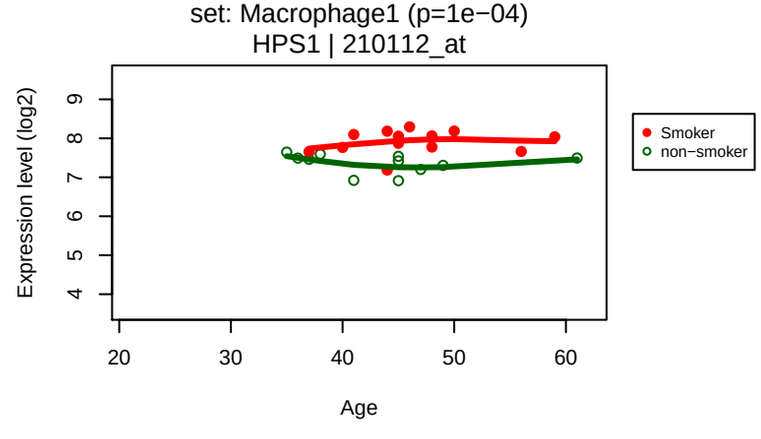

set: SmallAirway1 (p=0.7516)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker

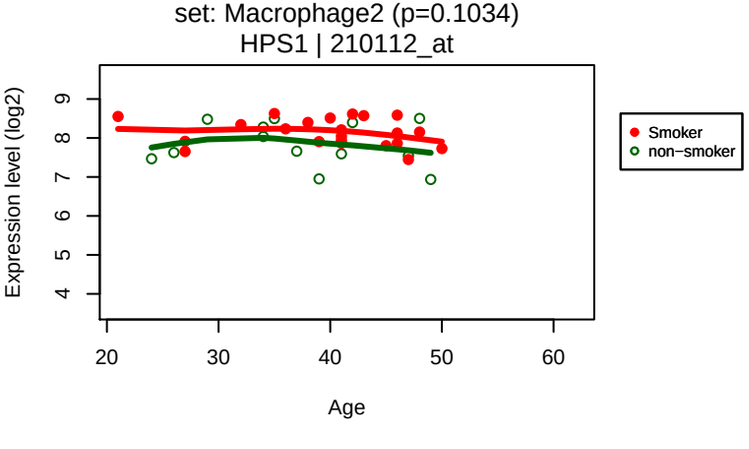

set: SmallAirway2 (p=0.0048)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker

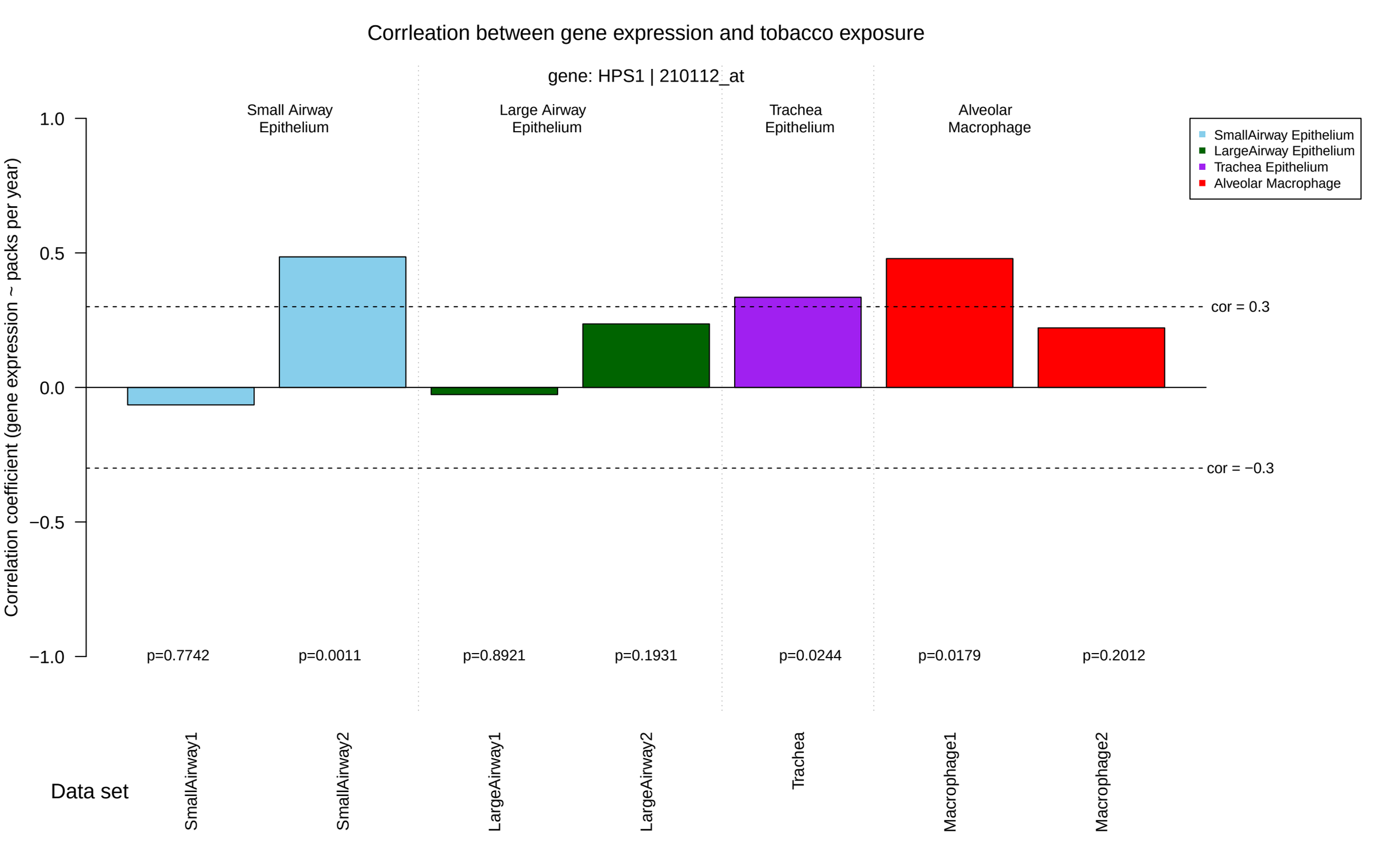

set: LargeAirway1 (p=0.9689)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1748)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0364)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=1e−04)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1034)
HPS1 | 210112_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS1 | 210112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7742
p=0.0011
p=0.8921
p=0.1931
p=0.0244
p=0.0179
p=0.2012
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2